Strings

Character
↳ 'A', 'a', - - - 1, 2, 3 - - 0
', , - - -

String : Seq of characters.
↳ char[] → ['a', 'd', 'i', 't', 'y', 'a']
↳ comparison (x)
↳ Concat (x)
↳ substring (x)
⋮

int → Integer
Char → Character
Float → Float
⋮ ⋮

char[] → String

String
↳ java.lang.String

```
Class String {
    //
}
```

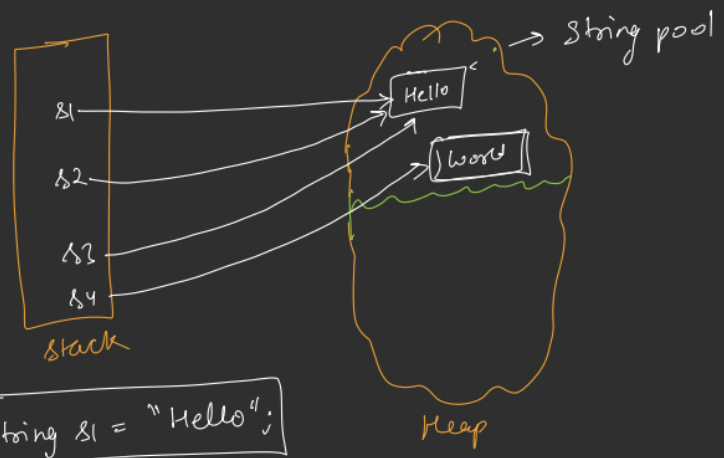
String s = new String();
String s = "Hello";

Integer i = 4;
↳ new Integer (4);
↳ Integer.valueOF(4);

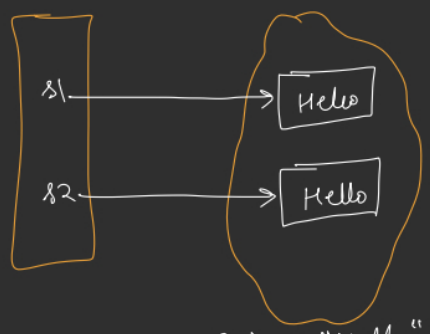
String
↳ Immutable Class.
↳ passwords / URL's / Hashes

String

- Literal
  - ↳ String Pool
- new




```
String s1 = "Hello";
String s2 = "Hello";
s3 = "Helo"   s4 = "world";
```




```
String s1 = new String("Hello");
String s2 = new String("Hello");
```

Golden Rule :

- ↳ only compile time constants go to String pool automatically.
- ↳ Runtime created string go to heap.

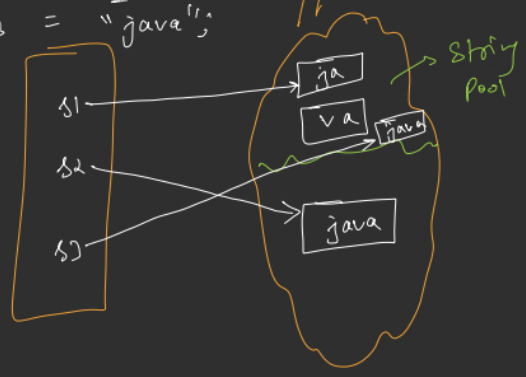
```
String s1 = "ja" + "va";
String s2 = java;
```

→ (s1 == s2) ? // true.

↳ compile-time

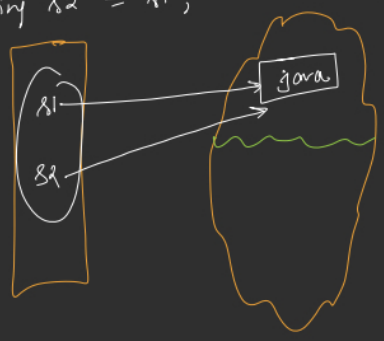
```
String s1 = "ja";
String s2 = s1 + "va";
String s3 = "java";
```

// (s2 == s3)

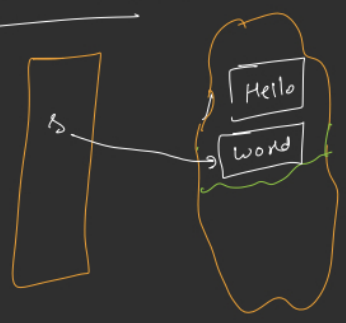



```
String s1 = "java";
String s2 = s1;
```

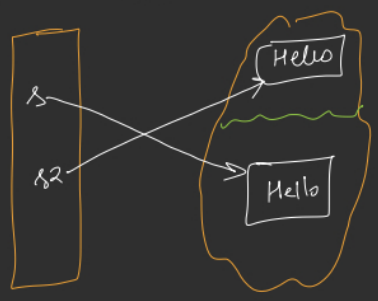
// (s1 == s2)
↳ true




```
String s = "Hello";
s = "world";
```



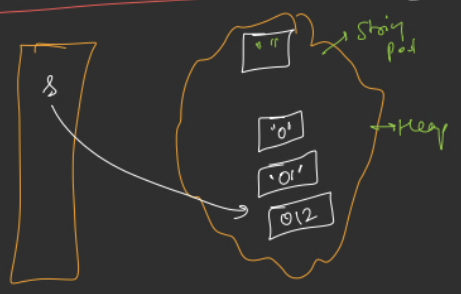

```
String s = new String("Hello");
```




```
String s2 = 'Hello';
```

```
String s = "";
for (i = 0; i < 5; i++) {
    s = s + i;
    System.out.println(s);
}
```

""
0
01
012
0123
01234




```
Public Final class String {
    Private Final byte[] values;
    Private Final byte coder;
    Private int hash;
}
```

s = "java";

coder = 0;

123 ← s.hashCode() → int

byte[] values

String s = "java";

values = | 74 | 97 | 118 | 97 |

String s = "java";

char[] values = | j | a | v | a |

2 bytes → UNICODE → ক

a → a? → ক

j = 2 byte
a = 2 byte
v = 2
a = 2

8 bytes

a – z
A – Z
0 – 9
, = ---

→ ASCII values

0 → 255

1 byte

byte[]

String s = "java";

byte[] | 74 | 97 | 118 | 97 |

1 1 1 1

= 4 byte

≈ 50%

s = "কাজী"

byte [coder];

- 0 → Latin1 → 1 byte
- 1 → UTF16 → 2 byte

s = " ";

→ UNICODE

s = "java"

values[] = | 74 | 97 | 118 | 97 |

coder = 0

s = "जावा";

values[] = | ? | ? | ? | ? |

coder = 1;

String Optimization (Immutable)

- String Pool
- Char[] → byte[]
- Caching the hashe values.